Virtual Reality in Musical Education

Daniel Grech
Institute of ICT
Malta College of Arts, Science and Technology
Paola, Malta
daniel.grech.g52665@mcast.edu.mt

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Abstract

Virtual Reality (otherwise known as VR) which can be labeled as a form of immersive multimedia, is the art of simulating ones physical presence for the user, threading the line between both reality and imagination. Virtual reality often utilises two of our senses, those being sight and hearing. This can be staggering for the music scene as anyone wishing to start learning an instrument but can not due to circumstances involving the likes of small space or noise pollution, could be approached with the use of virtual reality to simulate a realistic environment of a studio with instruments on a digital plane.

1 Introduction

Virtual Reality is a computer-generated simulation that creates an immersive experience for the user in a different world. Virtual reality technology has the ability to provide users with the feeling of immersion, allowing them to interact with and be able to experience virtual objects and the environment they are placed in, as if they were actually real. In music education, virtual reality could be utilised to enhance traditional teaching methods and be able to provide its users with new ways to engage in music. Virtual reality can provide a unique learning experience that can encourage exploration for its users. It can allow users to visualize themselves playing an instrument like the drums in a studio where they are more commonly found.

The purpose of this study is to investigate the applicability of virtual reality in music education, and test how effective virtual reality can be to assist users wishing to learn any instrument. The primary dependent variable is to determine learning outcomes; the extent to which whether users are improving on their musical skills after using VR technology. The independent variable would be the use of VR technologies. Hence the hypothesis of this study is to determine if people learn basic music education using solely virtual reality technologies, and if they able to compete with people learning basic music education physically.

Given the above hypothesis, 2 research questions were identified:

1. How do users that learnt music solely from VR technologies compare to users that have a basic background in music?

2. Are using VR technologies in music education ideal learning tools?

2 Literature Review

2.1 VR and Multimedia in music education

Multimedia and virtual reality have the potential to revolutionize the field of music education by providing its users with both immersive and engagingly learning experiences. Multimedia and virtual reality are currently the trend for assisting in teaching, with recent studies showing that students who utilized a form of optimized multimedia has shown a significant improvement in students' scores, and have also led to better quality of work [5].

One area where virtual reality has shown promise in musical education is in the improvement of music performance skills. Studies have shown that virtual-reality based training has shown benefits and improvements in piano performing skills in novice pianists. As shown in Table 1, 89 percent of students have shown a significant amount of progress in mastering musical literacy, 83 percent of students have developed new skills in knowing how to read sheet music, and 70 percent of students have shown basic performance

Skills	Percent
Mastered musical literacy	89
Able to read sheet music	83
Basic performance skills	70

Table 1: Students' musical improvements when using VR technologies.

skills when presented with a real piano, compared to a virtual one.

2.2 Benefits of music education

Music education had numerous benefits for individuals. Music education can improve people's cognitive, social and emotional development. Research has shown consistently that music education can have a great impact on ones academic achievements, social skills, and emotional well-being. Cognitively, music education helps by improving memory and lengthen attention span. According to studies, students who study music normally show increased levels of cognitive development compared to students who do not study music [4]. Socially, music education benefits students by giving them improved communication skills, better teamwork and an increased self-esteem. Studies show that students who participate in music education have significantly increased social skills and are normally more empathetic towards others when compared to students not engaging in music education [2]. Lastly, music education emotionally helps students by giving them increased self-expression and better emotional regulation. Studies show that students who engage in music education have more control over their emotional regulation and are more likely to express themselves in a healthier fashion when compared to students who do not participate in music education [3].

2.3 Music education for home-stuck individuals

Virtual reality technologies and other multimedia technologies can also educate individuals that are stuck at home, primarily due to disease or dysfunction in their limbs that makes it unable for them to walk around freely. During the COVID-19 pandemic, researchers have attempted to find ways for individuals wishing to learn a form of musical instrument to do so in the safety of their own home. Studies have shown that individuals wishing to learn music were able to do so with the use of virtual reality technologies, and multimedia technologies, and have even managed to retain some of their knowledge after leaving their homes and continuing their training [5].

2.4 What age should children start learning music

According to studies, learning music can help a child's cognitive, social, and emotional development. Some academics recommend beginning music instruction as early as possible because it has been discovered that early music education is particularly advantageous for kids. According to studies [1], children who began music lessons before the age of 7 showed significant improvements in cognitive abilities such as language processing and spatial-temporal skills compared to those who began after the age of 7. Further studies also found that children who started music lessons before the age of 6 showed greater improvements in reading, memory, and math skills compared to those who began later. However, some researchers suggest that starting music education too early may not be beneficial and could even be detrimental. In some studies, it was found that starting music lessons before the age of 7 did not necessarily lead to superior musical abilities in the long term and could result in a higher likelihood of quitting music lessons later on. The age at which children should begin learning music is a topic of ongoing debate, with some studies suggesting that starting music education at a young age can have significant benefits for cognitive and social development. However, it is important to consider individual differences and preferences when making decisions about when to begin music education. Further research is needed to determine the optimal age for children to begin music education. With the use of virtual reality technologies and the significant amount of benefits virtual reality technologies have, it can be an excellent alternative for children to learn music education using virtual reality technologies rather than having to spend more money and waste more space on physical instruments.

3 Research Methodology

This study aims to determine whether individuals that have played on a drum-kit virtually made in Unity can compare to individuals that have played a physical drum-kit. For this to be done, a family-friend with a physical drum-kit in his house was contacted, and has given permission to do this test. Figure 1 displays the procedure of the tests done.

The first stage of testing to test VR compatibility in music education is to first set up both

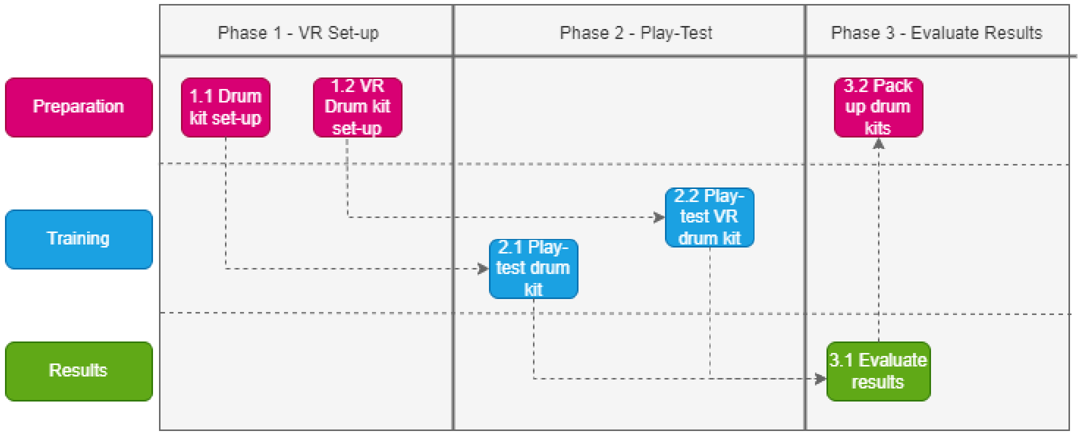

Figure 1: Research Pipeline Diagram

the virtual drum kit and the physical drum kit. Once both were prepared, four individuals were used for testing purposes; two will be playing the drums physically and the other two will be playing virtually. All four individuals were given a simple drum beat to follow, and in fifteen minutes have to try their best to replicate the beat given to them. Once the time has passed, all four individuals were asked to show their results of their studying and practicing of the drum beat by replicating it on the physical drum kit. Once they presented their efforts, both the virtual reality headset and drum kit were play tested by the other groups to get a feel. Finally, interviews happened and both the virtual reality head set and the physical drum kit were packed up.

4 Results

All four individuals in the tests were teenage males with no music experience whatsoever. The group testing on a physical drum kit shall be labelled as group 1, and the group testing on a virtual drum kit shall be labelled as group 2. Group 1 and group 2 were positioned in separate rooms, distant from each other as to minimise each team hearing the progress of the other team. After both groups did their time and tested their knowledge on the physical drums, both groups have managed to play the simple beat perfectly. After their tests, both groups swapped drum kits as to get a feel of how they work. Once that was complete, interviews were conducted on each individual to describe their experiences. They were first asked which drum kit do they prefer. All four of them stated that they would prefer the physical drum kit. Afterwards, they were asked if they would not mind using the virtual drum kit as a replacement if they could not have access to a physical drum kit, and all have responded with yes. Overall, all the participants preferred using a physical drum kit, but would not be opposed to using a virtual drum kit. These results indicate that both groups managed to learn how to play a simple beat even though they were using two different methods to play the drums, therefore this means that a virtual drum kit may be a possible solution for individuals who want to learn how to play the drums. This does however bring up a new hypothesis, that being if it is possible for individuals who have only used a virtual drum kit be able to compete with individuals who have experience with playing the drums.

5 Conclusion

The study overall was to test whether virtual reality instruments could compete with physical instruments, and the tests above have shown an indication that it might be possible for virtual reality instruments to compete with physical instruments. This is helpful for anyone stuck at home due to medical reasons or can not get instruments due to causing too much noise. The hypothesis for this study is to test whether virtual reality instruments can compete with physical instruments, and the results indicate that this may hold true. This study could lead to future studies to determine whether individuals solely play testing on virtual drum kits are able to compete with experienced drummers, to determine if virtual reality instruments are able to be a replacement from physical instruments.

6 References

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